

Notes: Bethune, Choi & Wright (RES, 2020)

Frictional Goods Markets: Theory and Applications

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1 Big picture

- **Question.** How do inflation, information, and credit conditions affect prices, markups, welfare, and the use of money in goods markets with explicit trading frictions?
- **Core idea.** The paper builds a dynamic monetary general equilibrium model in which some buyers are informed and can direct search toward attractive sellers, while other buyers are uninformed and meet sellers randomly.
- **Motivation.** Earlier search-and-bargaining monetary models often implied large welfare costs of inflation, while competitive-search models implied much smaller costs. The paper shows that both conclusions depend on goods-market microstructure.
- **Two search environments.** The first model has *partially directed search*: informed buyers, or locals, know all posted terms and direct search; uninformed buyers, or tourists, randomly search. The second model has *noisy search*: buyers observe a limited number of offers and choose the best one.
- **Sellers.** Ex ante identical sellers can choose different posted terms in equilibrium. Some offer low prices and high quantities to attract informed buyers; others post less attractive terms to exploit uninformed buyers.
- **Money and credit.** Buyers carry real balances z and may use costly credit d . Inflation taxes money balances, but cash and credit are substitutes.
- **Main quantitative result.** The welfare cost of 10 percent inflation is much smaller than in pure random-search bargaining models. In the baseline calibration it is about 0.8 percent of consumption in Model 1 and approximately zero in Model 2.
- **Nonstandard inflation benefit.** Inflation can have a second-best benefit by taxing cash purchases at high-price sellers that profit from uninformed buyers. It reduces the inefficient tourist-shop segment of the market.
- **Information result.** More information can raise welfare by directing buyers toward better terms and shrinking the high-price segment. But the effect depends on how information changes market composition, money demand, and markups.
- **Credit result.** Easier credit is not unambiguously good. It substitutes for money, changes entry and pricing, and can reduce welfare in the calibrated monetary economy.
- **Takeaway.** The welfare cost of inflation is not a single number produced by “money demand.” It depends on whether goods markets are random-search, directed-search, or mixtures of both, and on whether buyers can use credit and information.

2 Incremental contribution of the paper

2.1 What the paper adds relative to the monetary search literature

- Earlier New Monetarist work provides tractable models in which money is valuable because buyers and sellers trade in decentralized markets with matching frictions.
- Search-and-bargaining models, such as Lagos–Wright style environments, often imply large welfare costs of inflation because inflation directly distorts buyers’ ability to trade.
- Competitive-search or directed-search models, such as Rocheteau–Wright style environments, often imply smaller welfare costs because prices and trading terms adjust through posted contracts and directed entry.
- Bethune, Choi, and Wright do not choose one microstructure over the other. They build a framework that nests random search and directed search as limiting cases and studies intermediate cases with heterogeneous information.

Interpretation: The incremental contribution is not merely adding another calibration. It shows that the welfare cost of inflation is a function of market microstructure: who observes which prices, how sellers post terms, whether buyers can direct search, and whether credit substitutes for money.

2.2 What the paper adds relative to price-dispersion and information-search models

- The paper imports ideas from Diamond, Burdett–Judd, and Lester into a dynamic monetary economy.
- It formalizes Diamond’s idea that some consumers are tourists with little information while others are locals with better price information.
- Unlike static price-dispersion models, the model has money holdings, inflation, credit, entry, and welfare in general equilibrium.
- Sellers’ price dispersion is not imposed. It arises endogenously because sellers target different buyer groups.

Interpretation: The paper connects a classic industrial-organization problem—how price dispersion emerges from consumer information frictions—to a macro-monetary problem: how inflation and credit affect welfare.

2.3 What the paper adds relative to standard welfare-cost-of-inflation calculations

- Standard reduced-form monetary models usually treat inflation as a pure distortion of cash holdings.
- Search-and-bargaining models often produce high costs because inflation directly reduces decentralized-market trade.
- This paper shows that inflation can also have a benefit: it taxes transactions at high-price sellers that inefficiently exploit uninformed buyers.
- Therefore, the welfare cost of inflation can be low not because inflation is harmless, but because some of its distortions offset preexisting search inefficiencies.

Interpretation: The contribution is a second-best logic. Inflation is costly because it taxes money, but it can also reduce inefficient trade patterns created by information frictions.

2.4 What the paper adds relative to credit-and-money models

- The model allows money and credit to coexist, with credit costly to use.
- Credit eliminates a nuisance indeterminacy in pure-currency posted-price economies and lets the authors discipline the model using money and credit demand data.
- The credit margin matters for inflation: when buyers can substitute credit for cash, inflation affects both the payment mix and the composition of sellers.
- The paper shows that improving credit conditions can lower welfare in a monetary economy, because easier credit can interact poorly with market composition and cash holdings.

Interpretation: The model highlights a subtle point: credit is not simply a way to avoid monetary frictions. It changes the entire equilibrium allocation, including seller entry, search, markups, and money demand.

2.5 Bottom-line incremental contribution

- The paper provides a unified monetary-search framework that nests random search, directed search, partially informed buyers, noisy search, posted prices, bargaining, money, credit, and endogenous information acquisition.
- It uses this framework to reconcile why previous papers found very different welfare costs of inflation.
- It introduces a new policy message: inflation can be less costly than expected when it reduces rents earned by high-price sellers who exploit uninformed buyers.
- It produces new comparative statics for information and credit, showing that better information and better credit are not purely mechanical welfare improvements.

3 Model primitives and measurement

3.1 Agents and markets

- The environment follows a New Monetarist structure with two stages in each period: a centralized market (CM) and a decentralized market (DM).
- In the CM, agents can work, consume, rebalance money, and repay or settle financial positions.
- In the DM, buyers want a specialized good and sellers can produce it. Search and information frictions make trade non-Walrasian.
- Buyers enter the DM with money balances z and possibly use credit d .
- Sellers post terms of trade, including price p_j , quantity or quality q_j , and market tightness n_j for submarket j .

Interpretation: The CM gives tractability and allows money balances to be chosen, while the DM is where search, price posting, bargaining, credit, and inflation matter.

3.2 Buyer types: locals and tourists

- A fraction λ of buyers are informed and called locals.
- Locals observe all posted terms and can direct search toward the submarket offering the best expected payoff.
- The remaining fraction $1 - \lambda$ are uninformed and called tourists.
- Tourists do not observe terms before search and randomly meet sellers.
- The probability that an uninformed buyer ends up at a tourist shop or local shop depends on the relative mass of sellers in each submarket.

Interpretation: The local-tourist distinction is the model's information friction. It is also the source of endogenous price dispersion.

3.3 Submarkets and seller types

There can be two submarkets:

- L : local shops, which post attractive terms to attract informed buyers;
- T : tourist shops, which post less attractive terms and exploit uninformed buyers.

The probability that a buyer is in submarket j is summarized by ω_j , and market tightness is n_j .

$$\omega_T = \frac{(1 - \lambda)N_T}{N_L + N_T}, \quad \omega_L = \frac{N_L + \lambda N_T}{N_L + N_T}. \quad (1)$$

Interpretation: Tourist shops can coexist with local shops because uninformed buyers may accidentally visit them. This gives sellers an incentive to post high prices even though locals would avoid those shops.

3.4 Matching technology

Within each submarket, sellers and buyers meet according to a constant-returns matching technology. For sellers, the probability of meeting a buyer is $\alpha(n)$, and for buyers, the probability of meeting a seller is

$$\frac{\alpha(n)}{n}. \quad (2)$$

- The function $\alpha(n)$ is increasing in market tightness n .
- The buyer meeting probability $\alpha(n)/n$ is decreasing in n .
- Entry by sellers affects market tightness and hence matching probabilities.

Interpretation: More sellers make buyers more likely to trade but reduce the probability that any individual seller meets a buyer. This is the standard competitive-search tradeoff.

3.5 Money, credit, and inflation

- Buyers hold real money balances z when entering the DM.
- The nominal interest rate i captures the opportunity cost of carrying money.
- Inflation raises i through the Fisher logic and therefore taxes money balances.
- Buyers can use credit d , but credit has transaction cost $\gamma(d)$.
- Credit substitutes for money but is not frictionless.

Interpretation: The model does not treat money as a shortcut in utility. Money is useful because some DM trades require liquid balances, and inflation changes the cost of those balances.

3.6 Calibration targets

The calibration uses targets related to money demand, credit demand, markups, and price dispersion:

- average money holdings relative to GDP;
- average consumer credit relative to GDP;
- the relationship between money/credit demand and the short-term nominal interest rate;
- retail markup levels;
- price dispersion across retailers.

Interpretation: The model is disciplined by both payment behavior and goods-market pricing behavior. This is crucial because welfare depends on both the monetary distortion and the search/markup distortion.

4 Main model and equations

4.1 Value of money balances

Let $V(z)$ denote the DM value of entering with real balances z . Monetary equilibrium requires that the marginal value of money satisfies

$$1 + i = V'(z). \quad (3)$$

- If i rises, holding money becomes more expensive.
- In equilibrium, buyers reduce z when inflation or the nominal interest rate rises.
- Lower z affects not only quantities but also prices, credit use, and submarket composition.

Interpretation: This Euler condition is the monetary backbone of the model: inflation reduces money holdings by raising the opportunity cost of liquidity.

4.2 Seller free entry and pricing

Sellers enter a submarket until expected profit equals entry cost k . A seller chooses terms to maximize expected surplus subject to buyers' participation and bargaining/posting restrictions.

A generic submarket payoff expression is of the form

$$\Omega = \sum_j \omega_j \left\{ \frac{\alpha(n_j)}{n_j} [u(q_j) - q_j - \gamma(p_j - z)] - \frac{k}{n_j} \right\}. \quad (4)$$

Interpretation: Welfare combines consumption utility, production cost, credit transaction cost, and entry cost. A submarket can be inefficient either because quantities are distorted or because too many/few sellers enter.

4.3 Local and tourist shops

In equilibria with both local and tourist shops:

- local shops offer lower prices and higher quantities, $p_L < p_T$ and $q_L > q_T$;
- local shops have tighter terms from sellers' viewpoint because they face more informed demand;
- tourist shops can survive because uninformed buyers do not always avoid them.

Interpretation: This is the mechanism behind endogenous price dispersion: identical sellers behave differently because buyers differ in information.

4.4 Fixed information parameter λ

The first model treats λ as exogenous.

- If $\lambda = 0$, all buyers are uninformed and search is random.
- If $\lambda = 1$, all buyers are informed and search is directed.
- For intermediate λ , both information and random search matter.

Interpretation: This setup nests pure random search and pure directed search and shows how welfare changes continuously between them.

4.5 Endogenous information acquisition

The next model lets buyers choose whether to become informed by paying an information cost $s(\lambda)$.

Choose λ until marginal information benefit equals marginal information cost. (5)

- Higher λ helps buyers avoid tourist shops.
- Higher λ changes seller entry and the distribution of market tightness.
- Higher λ can reduce cash holdings because buyers are less likely to face unfavorable trades.

Interpretation: Information is endogenous, so policy changes like inflation or credit-market improvements can change not only money demand but also how informed buyers become.

4.6 Noisy search

The noisy-search version follows Burdett–Judd logic:

- some buyers observe only one seller offer;
- others observe more than one offer and choose the best;
- sellers post a distribution of terms instead of two discrete submarkets.

Interpretation: The noisy-search model shows that the main results are not special to a two-submarket local-tourist structure. Price dispersion and inflation effects also arise when buyers sample different numbers of offers.

5 Theoretical results

5.1 Existence and uniqueness with fixed information

Lemma 1. With fixed λ , equilibrium with $z = 0$ exists and is unique.

Lemma 2. In monetary equilibrium, $z < \hat{p}$ and $V''(z) < 0$ for relevant money balances.

Proposition 1. With fixed λ , monetary equilibrium with $z > 0$ exists and is unique if i is not too large. If local and tourist shops coexist, local shops offer better terms than tourist shops.

Interpretation: The model has a well-defined monetary equilibrium near empirically relevant inflation rates. The equilibrium features price dispersion when both locals and tourists are present.

5.2 Inflation with fixed information

The model studies how i affects z , p_j , q_j , n_j , and d_j .

- Higher i reduces money holdings z .
- Lower z raises reliance on credit in some trades.
- Higher i can change the composition of sellers by affecting tourist-shop profitability.
- Inflation can raise or lower average prices and markups depending on how seller composition changes.

Proposition 2. With fixed λ , the Friedman rule $i = 0$ is optimal near the pure random-search and pure directed-search extremes, but positive inflation can be optimal for intermediate information levels.

Interpretation: Positive inflation can be second-best because it reduces inefficient surplus captured by high-price tourist shops.

5.3 Endogenous information results

Proposition 3. Monetary equilibrium with endogenous $\lambda \in (0, 1)$ exists and is unique under suitable conditions. If local and tourist shops coexist, the comparison between local and tourist terms remains as before.

Proposition 4. With endogenous information, positive inflation can be optimal when the marginal cost of information is low. Credit improvements can raise or lower welfare depending on how they interact with money and information.

Interpretation: Inflation and credit policy affect the equilibrium amount of information acquisition. This feedback is absent in models with exogenous search technology.

5.4 Noisy-search results

Lemma 3. In noisy search, if enough buyers observe multiple offers, all sellers post the low return. Otherwise sellers mix over posted returns.

Proposition 5. With fixed λ , a stationary monetary equilibrium with noisy search exists and is unique under the paper's conditions.

Proposition 6. With noisy search and fixed λ , inflation and credit affect the posted-return distribution and money balances in ways analogous to the local-tourist model.

Propositions 7–8. With endogenous noisy search, equilibrium exists and comparative statics continue to show that inflation, credit, and information interact.

Interpretation: The model's main economic forces are robust to changing the information structure from local-tourist search to noisy search.

6 Quantitative applications and policy experiments

6.1 Calibration

- The paper calibrates two baseline models.
- Model 1 has relatively high bargaining power parameter θ and a moderate information level λ .
- Model 2 has different bargaining power and information parameters, generating high markups at tourist shops but fewer visits to them.
- The calibration matches money demand, credit demand, retail markup, and price-dispersion moments.

Interpretation: The quantitative results depend critically on matching both monetary moments and price-dispersion moments. The paper explicitly tries to discipline both sides.

6.2 Welfare cost of inflation

- The benchmark experiment computes the welfare cost of 10 percent inflation.
- In Model 1, the cost is about 0.8 percent of consumption.
- In Model 2, the cost is approximately zero when welfare is measured relative to the natural optimum.

- These costs are far below pure random-search bargaining estimates and closer to competitive-search estimates.

Interpretation: Inflation is not painless, but it is much less costly in this mixed-search environment because it partly offsets the distortion created by high-price tourist shops.

6.3 Hyperinflation

- For modest inflation, welfare costs can be small or even negative in calibrated environments.
- For hyperinflation, welfare costs become large and positive.
- The model eliminates currency at sufficiently high nominal interest rates.

Interpretation: The second-best benefit of inflation is limited. At high inflation, the standard monetary distortion dominates.

6.4 Information policy

- More information means a higher λ .
- Higher λ can shrink tourist shops and reduce markups.
- In the calibration, increasing λ from zero to one can be worth several percent of consumption.
- Starting from the calibrated λ , eliminating tourist shops can also generate sizable gains.

Interpretation: Consumer information is valuable because it changes market composition, not merely because it gives individuals better search outcomes.

6.5 Credit conditions

- A lower credit-cost parameter C makes credit easier to use.
- Easier credit reduces reliance on cash, changing the role of inflation.
- In the calibrated monetary economy, lowering credit costs can reduce welfare.
- In a nonmonetary economy, tighter credit can also reduce welfare, so the sign of credit-policy effects depends on the presence of money.

Interpretation: Credit and cash are substitutes at the individual level but not perfect substitutes at the aggregate level, because money can generate inflation revenue rebates whereas credit uses real resources.

7 Identification and empirical discipline

7.1 Nature of evidence

- The paper is a theory and calibration paper, not a reduced-form causal empirical paper.
- The model is disciplined by matching observed money demand, credit demand, markups, and price dispersion.
- Comparative statics are theoretical predictions that can be connected to data on payment choice, price dispersion, information access, and inflation.

Interpretation: The credibility of the quantitative results comes from calibration to multiple margins rather than from a quasi-experimental design.

7.2 What the calibration identifies

- Money-demand data discipline the cost of holding cash and the curvature of DM utility.
- Credit-demand data discipline the cost of credit.
- Markup and price-dispersion data discipline the degree of information friction and bargaining power.
- The model then predicts welfare costs and comparative statics for inflation, information, and credit.

Interpretation: The moments are chosen to pin down the exact frictions that drive the welfare experiments.

7.3 Main limitations

- The model abstracts from many institutional details of retail markets and credit contracts.
- The information parameter λ is a reduced-form representation of consumer information.
- Calibration targets, especially markups and price dispersion, are difficult to measure cleanly.
- Welfare results depend on the model's equilibrium selection and assumptions about bargaining, posting, entry, and credit costs.

Interpretation: The model should be read as a disciplined theoretical laboratory, not as a complete empirical description of all goods-market transactions.

8 Tables and figures

8.1 Figure 1 and Figure 2: market structure and equilibrium regions

Read:

- Figure 1 shows the two-submarket structure: local shops attract informed buyers, while tourist shops are visited by uninformed buyers.
- A fraction λ of buyers are informed and direct search; the rest search randomly.
- Figure 2 shows monetary equilibrium regions in (λ, i) space.
- Region A_1 has both local and tourist shops, with $p_T > z > p_L$.
- Region A_2 has $p_T > p_L > z$, so buyers rely more on credit.
- Region A_3 eliminates tourist shops.

Economic message: The figures show why inflation can have nonstandard effects: changing i can move the economy across regions and change which seller types survive.

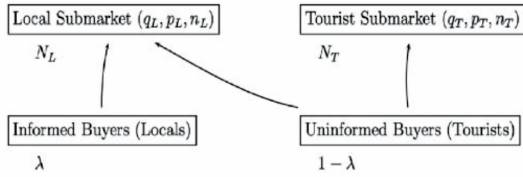


FIGURE 1

Decentralized market structure

Figure 1: decentralized market structure

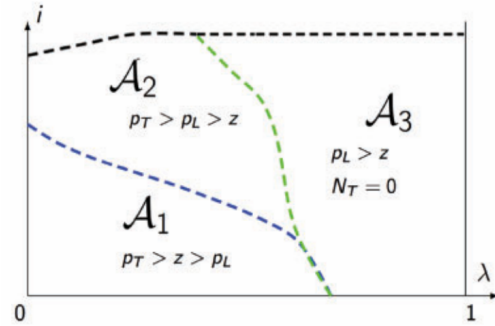


FIGURE 2

Monetary equilibrium regions in (λ, i) space

Figure 2: monetary equilibrium regions

8.2 Figure 3 and Table 1: calibration targets and calibrated parameters

Read:

- Figure 3 compares model-implied money and credit demand curves to historical data.
- The top-left panel fits money holdings as a percentage of GDP against the 3-month Treasury rate.
- The top-right panel fits consumer credit as a percentage of GDP.
- The bottom-left panel shows the distribution of DM markups.
- Table 1 reports calibrated parameters for Model 1 and Model 2, including utility curvature, credit cost, information cost, entry cost, and bargaining power.

Economic message: The paper calibrates the model to both payment behavior and price-dispersion behavior, which is why the quantitative welfare results differ from simpler monetary models.

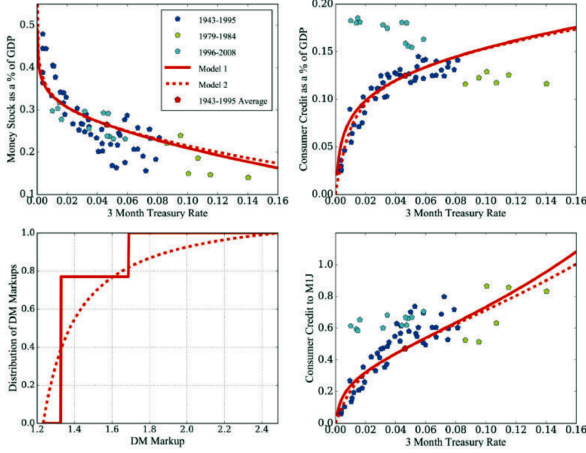


FIGURE 3
Estimated money and credit demand curves

Figure 3: money and credit demand curves

TABLE 1 Calibrated parameter values			
Description	Model 1	Model 2	Source/Target
DM utility curvature, b	0.67	0.58	$(z/Y, i)$ relationship
DM utility level, B	0.56	0.59	Avg. z/Y
Credit cost curvature, c	5.29	6.31	$(D/Y, i)$ relationship
Credit cost level, C	12.28	82.12	Avg. D/Y
Cost of information level, E	0.052	0.046	Retail markup 40%
Sellers' entry cost, k	0.020	0.014	Avg. markup 10%
Bargaining power, θ	0.72	0.31	Avg. price dispersion

Table 1: calibrated parameter values

8.3 Figure 4 and Figure 6: welfare cost of inflation

Read:

- Figure 4 compares the welfare cost of inflation across model specifications.
- Pure random-search bargaining-like specifications generate high costs.
- Adding entry, credit, and endogenous information greatly reduces the cost.
- Figure 6 extends the analysis to hyperinflation and shows that costs eventually become very large.

Economic message: Moderate inflation can have small welfare costs in the baseline model, but high inflation is still very costly. The model reconciles low moderate-inflation costs with large hyperinflation costs.

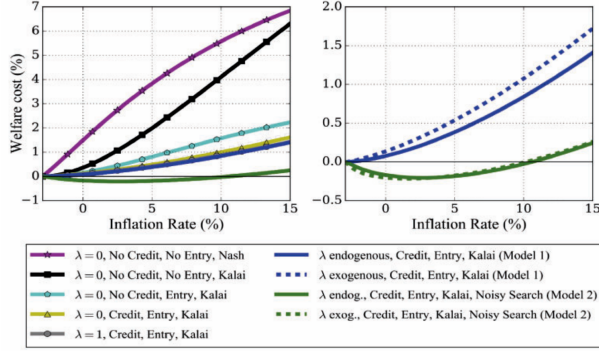


FIGURE 4
Cost of inflation: model comparisons

Figure 4: cost of inflation

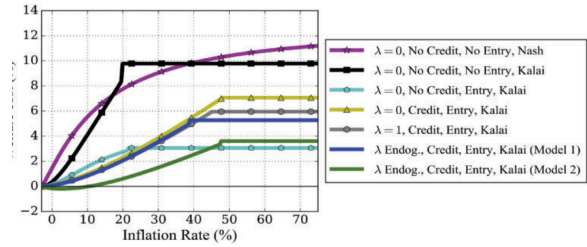


FIGURE 6
Cost of hyper-inflation: model comparisons

Figure 6: cost of hyperinflation

8.4 Figure 5: markups and price dispersion

Read:

- The left panel shows how inflation affects tourist and local markups in Model 1.
- The right panel shows how price/markup dispersion changes with inflation.
- Average DM markups can rise or fall depending on the distribution of trade across seller types.

Economic message: Inflation affects welfare partly through markups and price dispersion, not only through the quantity of money balances.

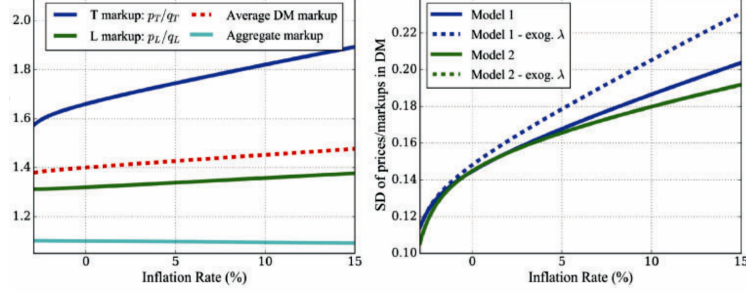


FIGURE 5
Effects of inflation on markups (in Model 1) and price/markup dispersion

8.5 Figure 7 and Figure 8: information and credit experiments

Read:

- Figure 7 shows welfare and money-holding effects of changing information costs, represented by changes in λ .
- More information can reduce tourist-shop distortions and raise welfare.
- Figure 8 shows the effect of changes in credit costs on welfare and money holdings.
- Better credit reduces cash holdings but does not necessarily improve welfare in the monetary economy.

Economic message: The policy effects of information and credit are general-equilibrium effects. They change the composition of sellers, the use of money, and the use of credit simultaneously.

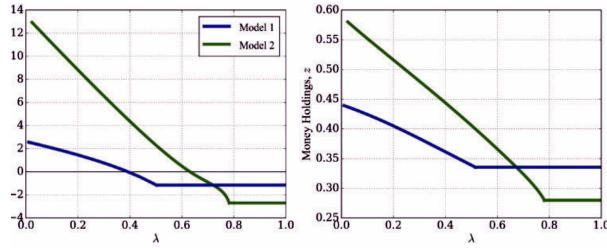


FIGURE 7
Effects of changes in information costs on Ω and z
Figure 7: effects of information costs

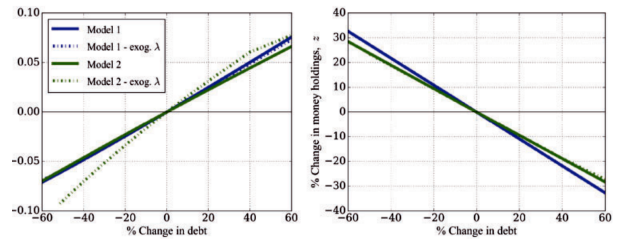


FIGURE 8
Effects of changes in debt induced by changes in the cost of credit
Figure 8: effects of credit costs

9 Short summary

- The paper studies dynamic monetary economies with frictional goods markets and heterogeneous buyer information.
- Some buyers are informed locals who direct search, while others are uninformed tourists who search randomly.
- Sellers endogenously sort into low-price local shops and high-price tourist shops.
- Money and credit coexist; inflation taxes money while credit is costly to use.
- The model nests random-search and directed-search environments.
- With fixed information, monetary equilibrium exists and is unique for moderate inflation.
- With endogenous information, inflation and credit also affect how informed buyers choose to become.

- In the calibration, the welfare cost of 10 percent inflation is much lower than in pure random-search bargaining models.
- Inflation can have a second-best benefit by reducing inefficient trade at high-price tourist shops.
- More information can be valuable by shrinking the tourist segment of the market.
- Easier credit can reduce welfare in a monetary economy because it changes money demand and market composition.
- The broad lesson is that goods-market microstructure is central for quantitative monetary policy.

10 Conclusion

10.1 Core conclusion

Bethune, Choi, and Wright show that the welfare effects of inflation, information, and credit depend critically on the microstructure of goods markets. When buyers differ in information and sellers post terms accordingly, inflation affects not only money holdings but also prices, markups, credit use, entry, and the composition of sellers.

10.2 Economic intuition

Inflation is a tax on money, but in this environment it is also a tax on a particular set of trades. High-price tourist shops exploit uninformed buyers and often rely on cash balances. Inflation reduces the value of carrying cash to those trades and can shrink the inefficient tourist-shop segment. This second-best benefit offsets part of the standard monetary cost.

10.3 Incremental contribution takeaway

The paper’s incremental contribution is to reconcile divergent welfare-cost estimates in the monetary literature by embedding heterogeneous information and goods-market search into a dynamic monetary model with credit. It shows that neither pure random search nor pure directed search is enough: the intermediate cases are where the most interesting monetary-policy effects arise.

10.4 Final takeaway

A calibrated monetary model should not treat the goods market as a black box. The welfare cost of inflation is jointly determined by money demand, credit substitution, seller entry, consumer information, and price dispersion. This paper provides a unified framework for studying those forces together.